Kelompok B 29, April 2026

MID EXAM CCI

Network

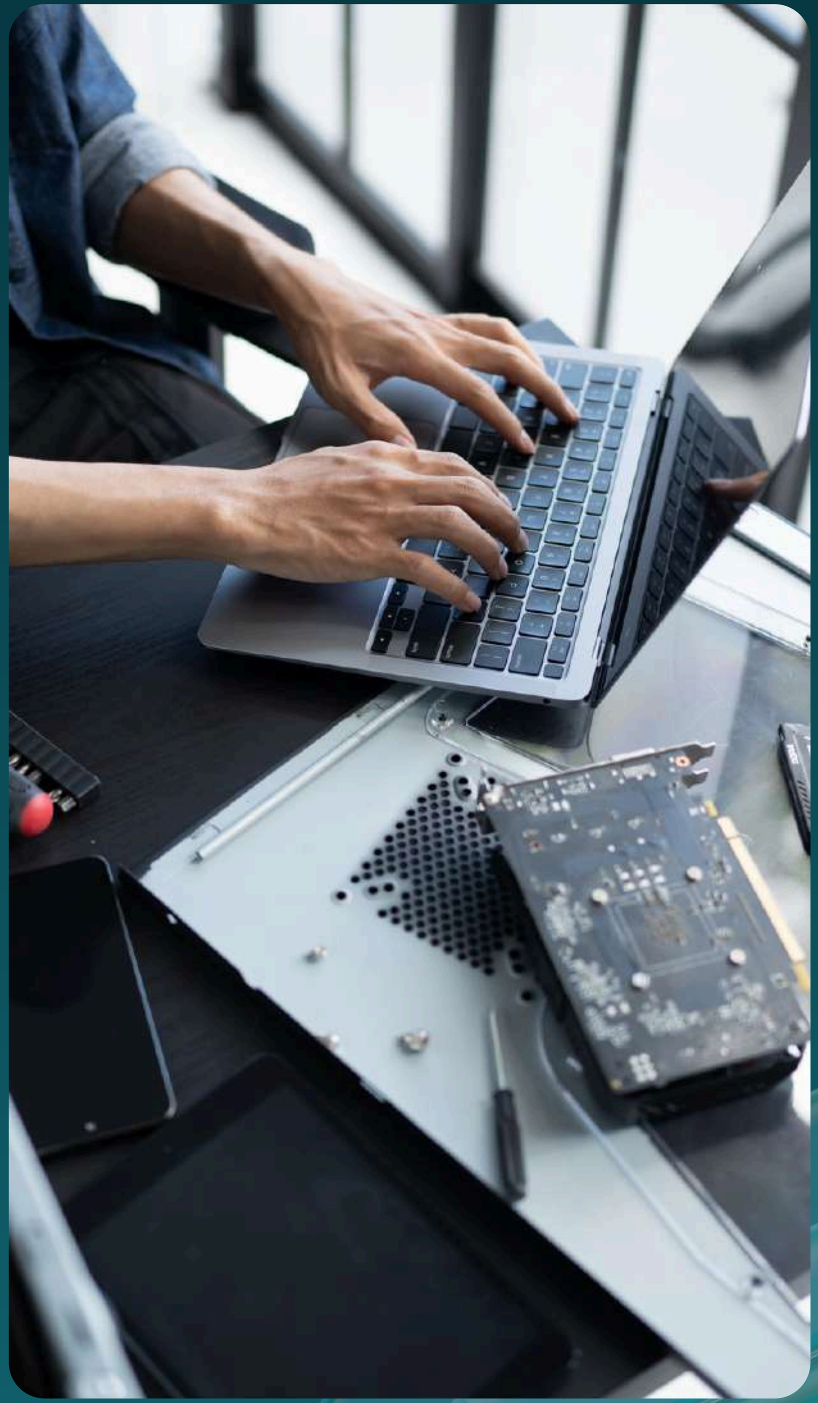
CENTRAL COMPUTER
IMPROVEMENT

Presented By: **Kelompok B**

Kelompok B

29, April 2026

Anggota Kelompok



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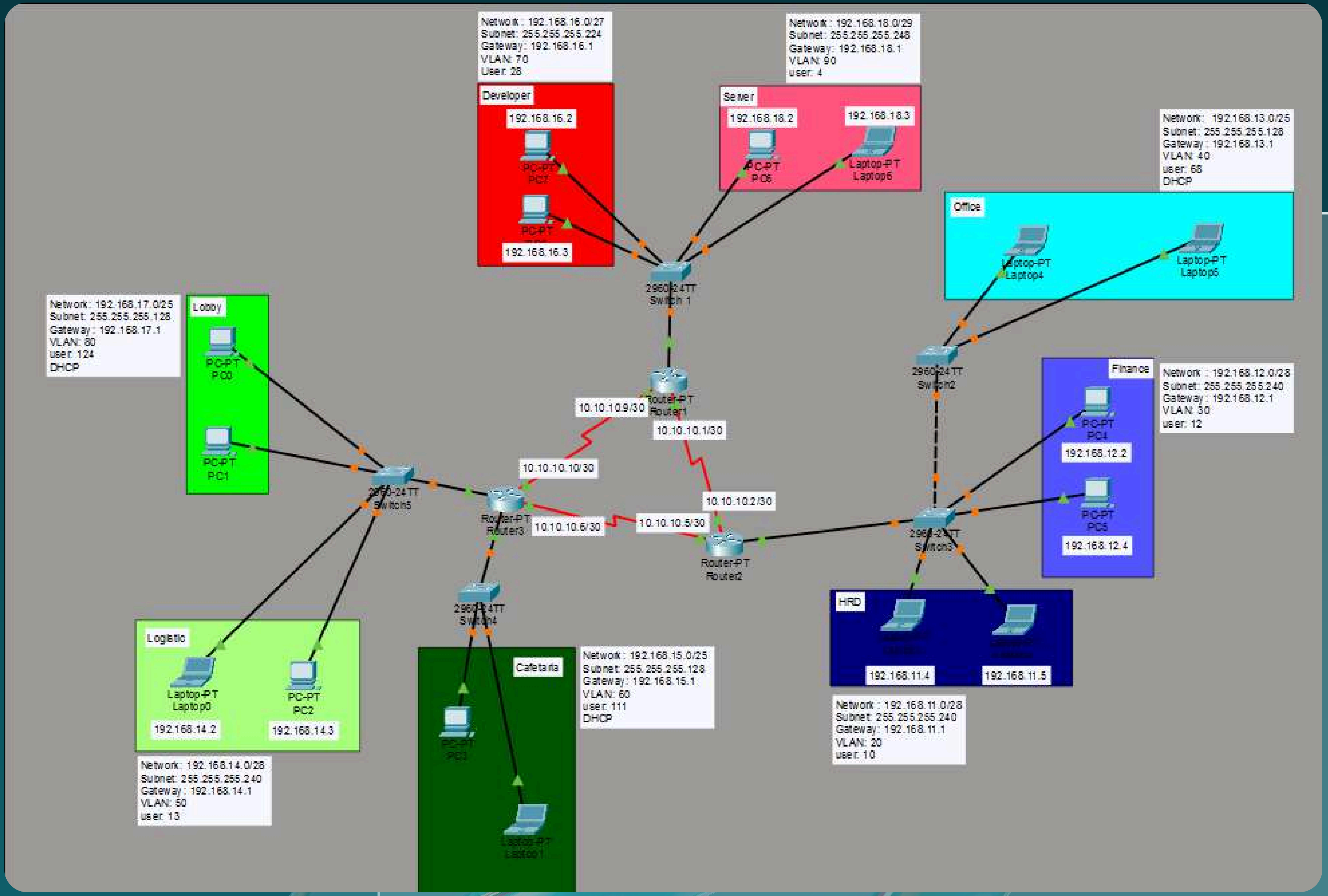
Topologi Jaringan



Konfigurasi Switch 1

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 70
Switch(config-vlan)#name developer
Switch(config-vlan)#vlan 90
Switch(config-vlan)#name server
Switch(config-vlan)#exit
Switch(config)#int range fa0/2 - 3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 70
Switch(config-if-range)#int range fa0/4 - 5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 90
Switch(config-if-range)#exit
Switch(config)#int fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#
```

Konfigurasi Switch 2

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 40
Switch(config-vlan)#name office
Switch(config-vlan)#exit
Switch(config)#int range fa0/2 - 3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 40
Switch(config-if-range)#exit
Switch(config)#int fa0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#
```

Konfigurasi Switch 3

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 30
Switch(config-vlan)#name finance
Switch(config-vlan)#vlan 40
Switch(config-vlan)#name office
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name HRD
Switch(config-vlan)#exit
Switch(config)#int range fa0/3 - 4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#int rnage fa0/5 - 6
                           ^
% Invalid input detected at '^' marker.

Switch(config-if-range)#int range fa0/5 - 6
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#int range fa0/1 - 2
Switch(config-if-range)#switchport mode trunk
Switch(config-if-range)#exit
Switch(config)#
```

Konfigurasi Switch 4

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 60
Switch(config-vlan)#name cafetaria
Switch(config-vlan)#exit
Switch(config)#int range fa0/2 - 3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 60
Switch(config-if-range)#int fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#
```

Konfigurasi Switch 5

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 50
Switch(config-vlan)#name logistic
Switch(config-vlan)#vlan 80
Switch(config-vlan)#name lobby
Switch(config-vlan)#exit
Switch(config)#int range fa0/2 - 3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switcport access vlan 50
                             ^
% Invalid input detected at '^' marker.

Switch(config-if-range)#switchport access vlan 50
Switch(config-if-range)#int range fa0/4 - 5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 80
Switch(config-if-range)#exit
Switch(config)#int fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#
```

Konfigurasi Router 1

```
Router(config)#int se2/0
Router(config-if)#no sh
Router(config-if)#ip add 10.10.10.1 255.255.255.252
Router(config-if)#int se3/0
Router(config-if)#ip add 10.10.10.9 255.255.255.252
Router(config-if)#no sh
```

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0.70
Router(config-subif)#encap
Router(config-subif)#encapsulation dot1Q 70
Router(config-subif)#ip add 192.168.16.1 255.255.255.224
Router(config-subif)#int fa0/0.90
Router(config-subif)#encap
Router(config-subif)#encapsulation dot1Q 90
Router(config-subif)#ip add 192.168.18.1 255.255.255.248
Router(config-subif)#exit
Router(config)#
```

Konfigurasi Router 2

```
Router(config)#int se2/0
Router(config-if)#ip add 10.10.10.5 255.255.255.252
Router(config-if)#int se3/0
Router(config-if)#ip add 10.10.10.2 255.255.255.252
Router(config-if)#no sh
```

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0.20
Router(config-subif)#encap
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip add 192.168.11.1 255.255.255.240
Router(config-subif)#int fa0/0.30
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip add 192.168.12.1 255.255.255.240
Router(config-subif)#int fa0/0.40
Router(config-subif)#encapsulation dot1Q 40
Router(config-subif)#ip add 192.168.13.1 255.255.255.128
Router(config-subif)#exit
Router(config)#int fa0/0
Router(config-if)#no sh
```

Konfigurasi Router 3

```
Router(config-if)#int se2/0
Router(config-if)#ip add 10.10.10.10 255.255.255.252
Router(config-if)#int se3/0
Router(config-if)#ip add 10.10.10.6 255.255.255.252
```

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0.60
Router(config-subif)#encap
Router(config-subif)#encapsulation dot1Q 60
Router(config-subif)#ip add 192.168.15.1 255.255.255.128
Router(config-subif)#int fa1/0.50
Router(config-subif)#encapsulation dot1Q 50
Router(config-subif)#ip add 192.168.14.1 255.255.255.240
Router(config-subif)#int fa1/0.80
Router(config-subif)#encapsulation dot1Q 80
Router(config-subif)#ip add 192.168.17.1 255.255.255.128
Router(config-subif)#exit
Router(config)#int fa0/0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%LINK-3-UPDOWN: Interface FastEthernet0/0.60, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.60, changed state to up

Router(config-if)#int fa1/0
Router(config-if)#no sh
```

Konfigurasi DHCP POOL

pada router yang terhubung dengan office (router 2)

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp pool DHCP_OFFICE
Router(dhcp-config)#network 192.168.13.0 255.255.255.128
Router(dhcp-config)#default-router 192.168.13.1
Router(dhcp-config)#
```

Konfigurasi DHCP POOL

pada router yang terhubung dengan cafetaria & lobby (router 3)

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp pool DHCP_CAFE
Router(dhcp-config)#network 192.168.15.0 255.255.255.128
Router(dhcp-config)#default router 192.168.15.1
                            ^
% Invalid input detected at '^' marker.

Router(dhcp-config)#default-router 192.168.15.1
Router(dhcp-config)#ip dhcp pool DHCP_LOBBY
Router(dhcp-config)#network 192.168.17.0 255.255.255.128
Router(dhcp-config)#default-router 192.168.17.1
Router(dhcp-config)#
```

Konfigurasi Static Route:

R1 (Router Atas):
-Static Route to Office Section.
-Static Route to Finance Section.
-Static Route to HRD Section.
-Static Route to Logistik Section.
-Static Route to Cafetaria Section.
-Static Route to Lobby Section.

R2 (Router Kanan):
-Static Route to Developer Section.
-Static Route to Server Section.
-Static Route to Logistik Section.
-Static Route to Cafetaria Section.
-Static Route to Lobby Section.

R3 (Router Kiri):
-Static Route to Developer Section.
-Static Route to Server Section.
-Static Route to Office Section.
-Static Route to Finance Section.
-Static Route to HRD Section.

R1:

```
Router(config)#ip route 192.168.13.0 255.255.255.128 10.10.10.2
Router(config)#ip route 192.168.12.0 255.255.255.240 10.10.10.2
Router(config)#ip route 192.168.11.0 255.255.255.240 10.10.10.2
Router(config)#
Router(config)#ip route 192.168.14.0 255.255.255.240 10.10.10.10
Router(config)#ip route 192.168.15.0 255.255.255.128 10.10.10.10
Router(config)#ip route 192.168.17.0 255.255.255.128 10.10.10.10
Router(config)#
```

R2:

```
Router(config)#ip route 192.168.16.0 255.255.255.224 10.10.10.1
Router(config)#ip route 192.168.18.0 255.255.255.248 10.10.10.1
Router(config)#
Router(config)#ip route 192.168.14.0 255.255.255.240 10.10.10.6
Router(config)#ip route 192.168.15.0 255.255.255.128 10.10.10.6
Router(config)#ip route 192.168.17.0 255.255.255.128 10.10.10.6
Router(config)#
```

R3:

```
Router(config)#ip route 192.168.16.0 255.255.255.224 10.10.10.9
Router(config)#ip route 192.168.18.0 255.255.255.248 10.10.10.9
Router(config)#
Router(config)#ip route 192.168.13.0 255.255.255.128 10.10.10.5
Router(config)#ip route 192.168.12.0 255.255.255.240 10.10.10.5
Router(config)#ip route 192.168.11.0 255.255.255.240 10.10.10.5
Router(config)#
```

Test- Ping:

verifikasi satu subnet (HRD -> HRD), beda route (HRD->Developer), inter vlan (HRD->Finance)

```
C:\>ping 192.168.11.5

Pinging 192.168.11.5 with 32 bytes of data:

Reply from 192.168.11.5: bytes=32 time<1ms TTL=128
Reply from 192.168.11.5: bytes=32 time<1ms TTL=128
Reply from 192.168.11.5: bytes=32 time<1ms TTL=128
Reply from 192.168.11.5: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.11.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.16.1

Pinging 192.168.16.1 with 32 bytes of data:

Reply from 192.168.16.1: bytes=32 time=25ms TTL=253
Reply from 192.168.16.1: bytes=32 time<1ms TTL=253
Reply from 192.168.16.1: bytes=32 time=1ms TTL=253
Reply from 192.168.16.1: bytes=32 time=1ms TTL=253

Ping statistics for 192.168.16.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 25ms, Average = 6ms

C:\>ping 192.168.12.1

Pinging 192.168.12.1 with 32 bytes of data:

Reply from 192.168.12.1: bytes=32 time<1ms TTL=255
Reply from 192.168.12.1: bytes=32 time<1ms TTL=255
Reply from 192.168.12.1: bytes=32 time<1ms TTL=255
Reply from 192.168.12.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.12.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Perbedaan Static Routing Dengan Dynamic Routing:

1. Jelaskan perbedaan antara Static Routing dan Dynamic Routing dalam hal konfigurasi, kelebihan, dan kekurangannya. Sertakan contoh penerapan masing-masing metode dalam jaringan ?

Static Routing

Static Routing adalah suatu config routing yang ditentukan secara manual oleh admin di setiap router dan kita harus mengetahui terlebih dahulu dari sebuah topologi jaringan dan alamat ip address yang ingin dituju serta next-hop juga.

Kelebihan Static Routing:
- Konfigurasi Yang Sederhana.
- Tidak Membebani CPU/Router.
- Lebih Aman.

Kekurangan Static Routing:
- Tidak Fleksibel.
- Diharuskan update secara manual ketika ada perubahan besar.
- Ribet jika diterapkan di jaringan skala menengah hingga besar.

Contoh Penerapan Static Routing:
- Jaringan Skala Kecil (Seperti Rumah, Lab, dan Sekolah Sederhana).

Dynamic Routing

Dynamic Routing adalah suatu config yang dibuat otomatis oleh router menggunakan suatu protokol.

Segi Konfigurasi:
- Konfigurasi Pada Dynamic Routing ini adalah bersifat otomatis. Jadi Router akan saling bertukar informasi menggunakan suatu protokol (seperti: RIP, OSPF, EIGRP dll), jadi kita tidak perlu isi satu- satu/manual seperti Static Routing.

Kelebihan Dynamic Routing:
- Otomatis Update Jika Adaa Perubahan.
- Lebih Cocok & Fleksibel Untuk Jaringan Skala Besar.
- Sangat Praktis, Jadi Tidak Perlu Setting & Config Manual Banyak.

Kekurangan Dynamic Routing:
- Konfigurasi Lebih Rumit.
- Memakai Resource Seperti (CPU,dan Bandwidth).

Contoh Penerapan Dynamic Routing:
- Jaringan Skala Menengah - Besar (Seperti ISP, Perusahaan Besar Atau Bercabang).

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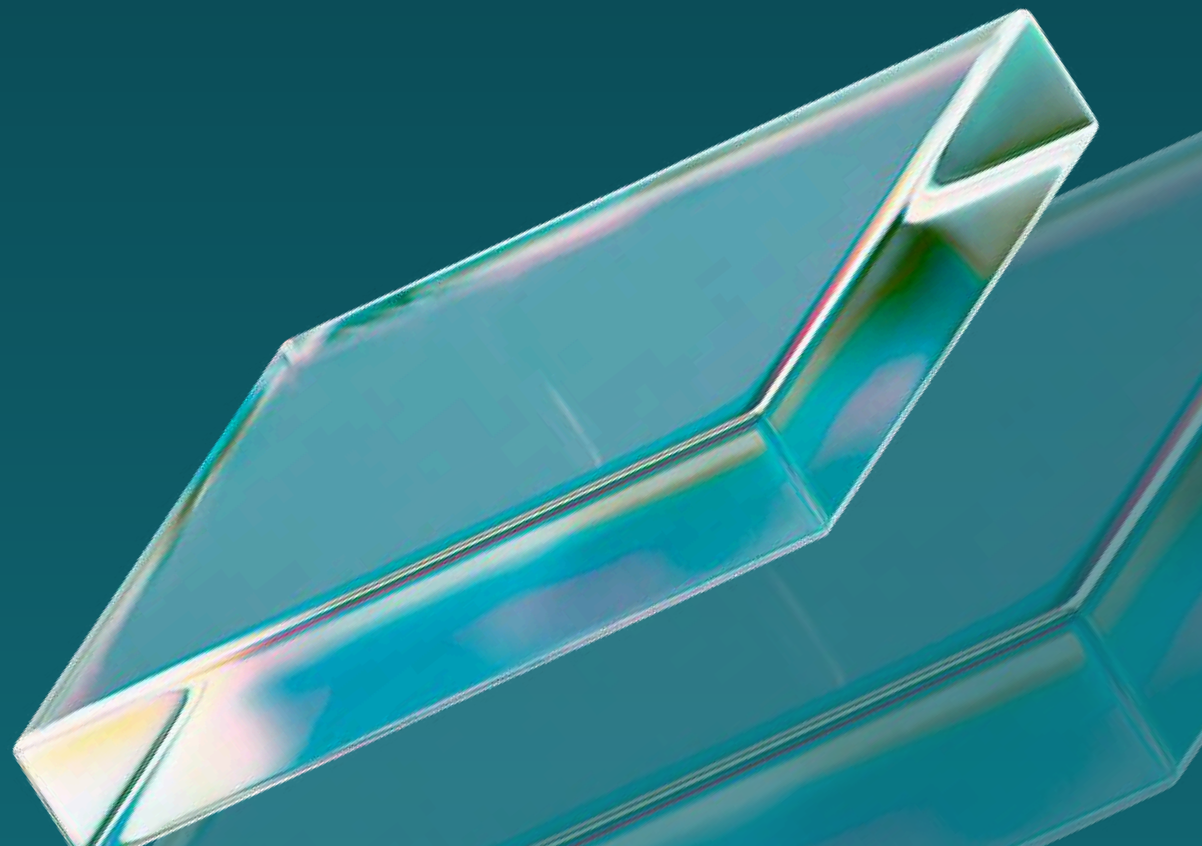
Thank You



Presented By: **Kelompok B**